

Target and stitched multiple-shooting estimate

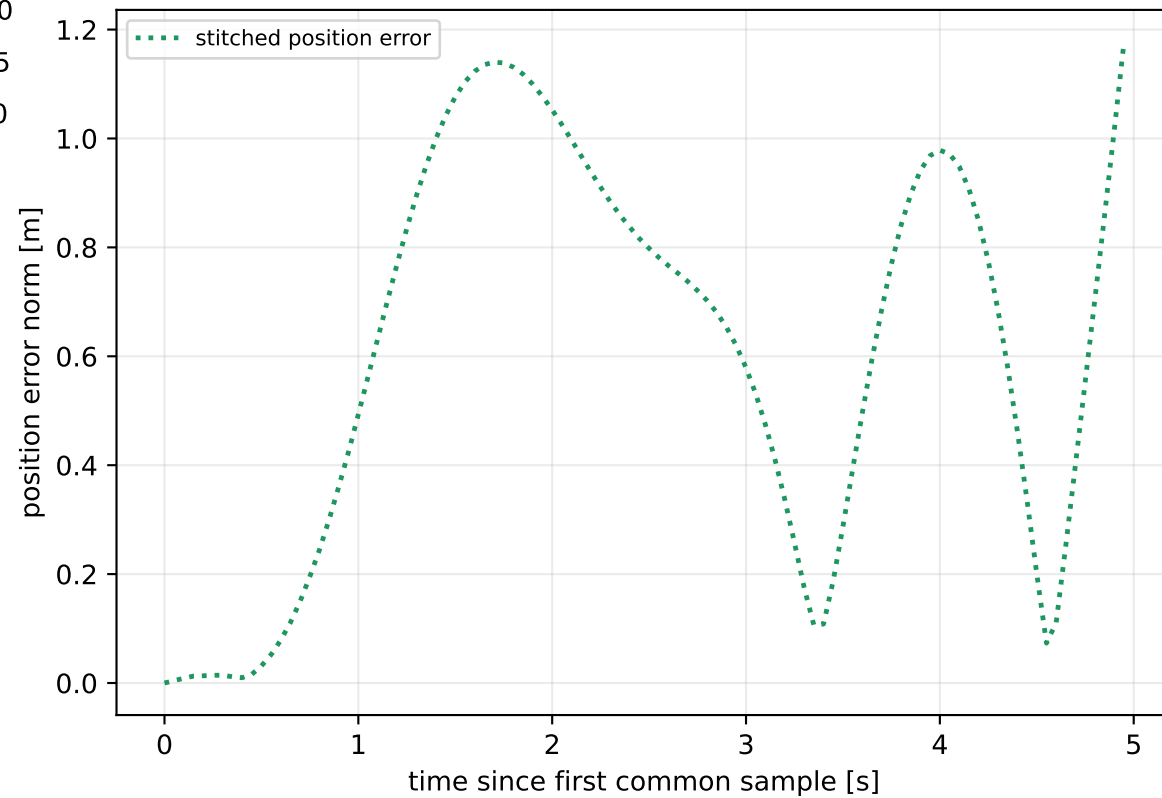
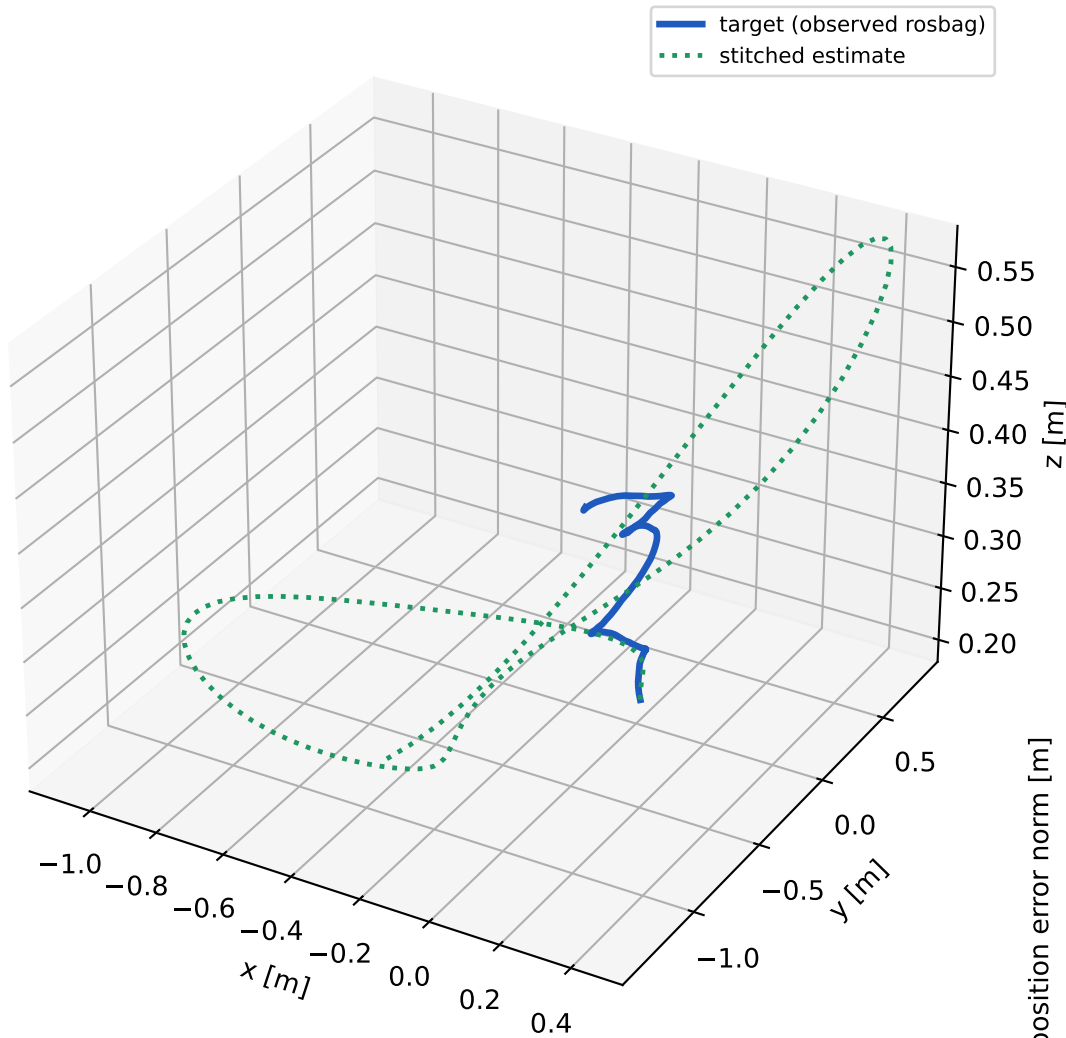
actual common support: 19.009–23.959 s
selected command delay: 0.0527255 s

metric	stitched
position RMSE [m]	0.70885
orientation RMSE [deg]	48.341
velocity RMSE [m/s]	1.4758
angular velocity RMSE [rad/s]	1.0275
specific force RMSE [m/s ²]	1.0747

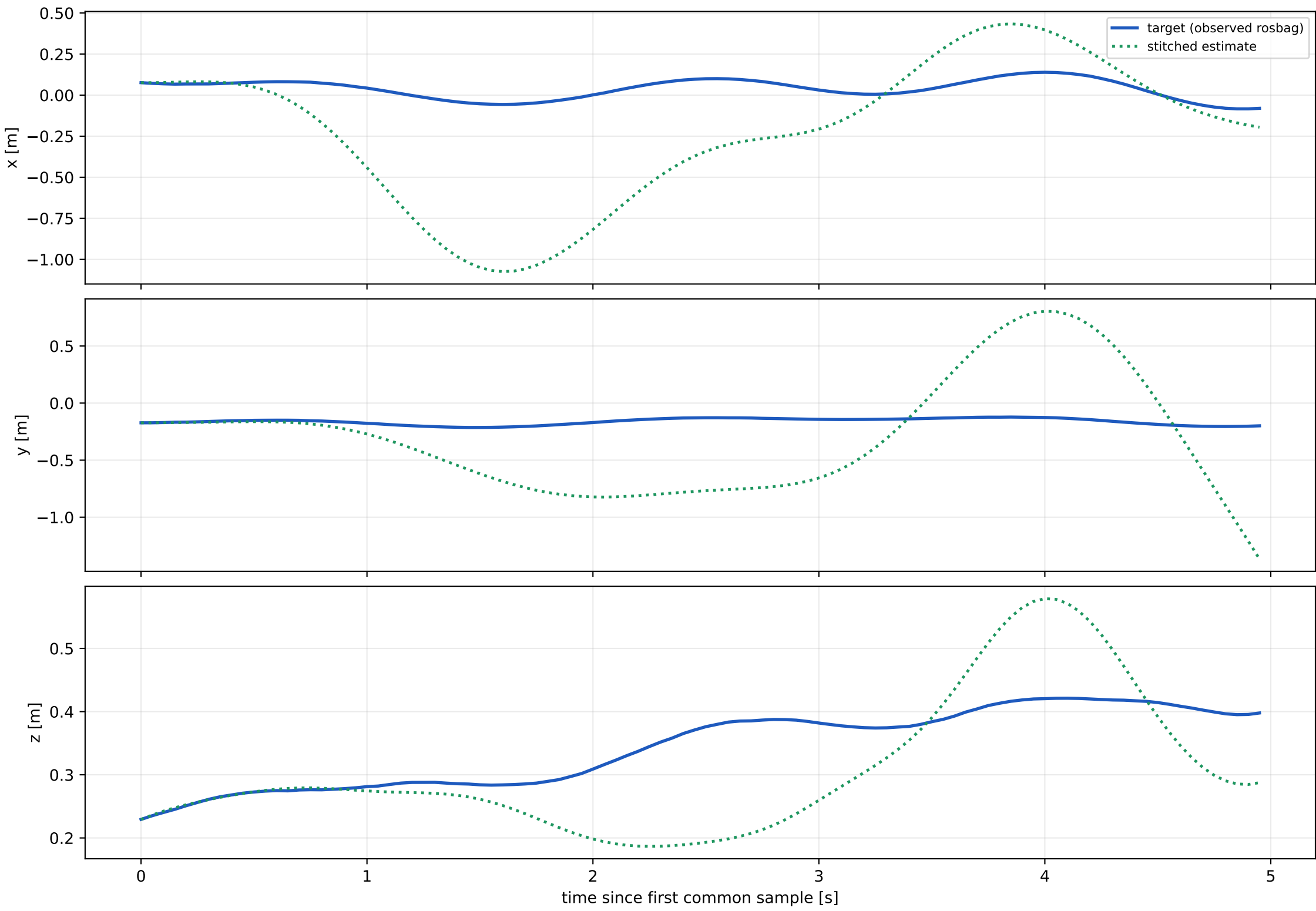
continuity max / tolerance:
0.0001343 / 0.001 (converged)

Velocity and IMU metrics are diagnostic only;
the fit uses sensor position and orientation.

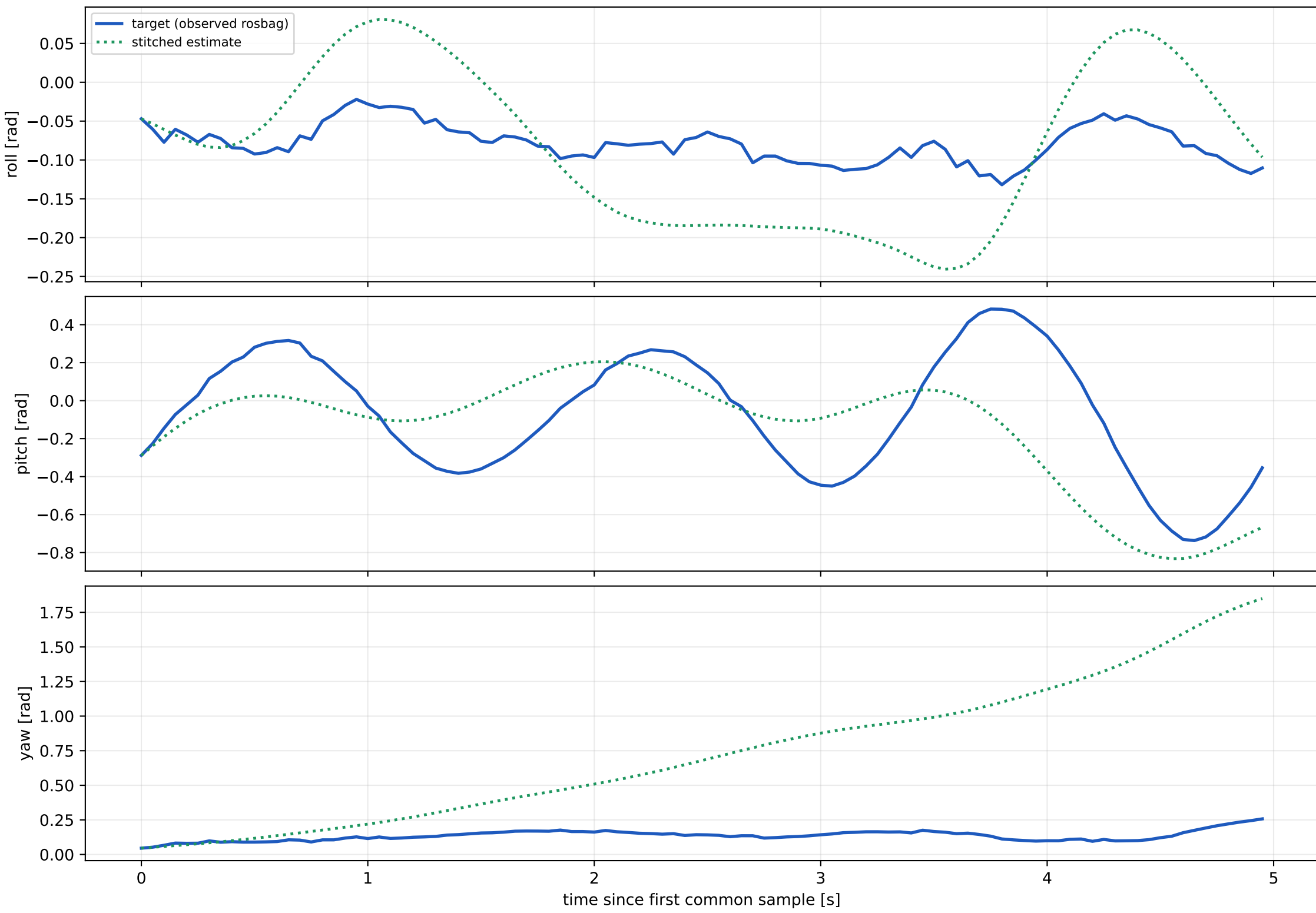
Stitched trajectory



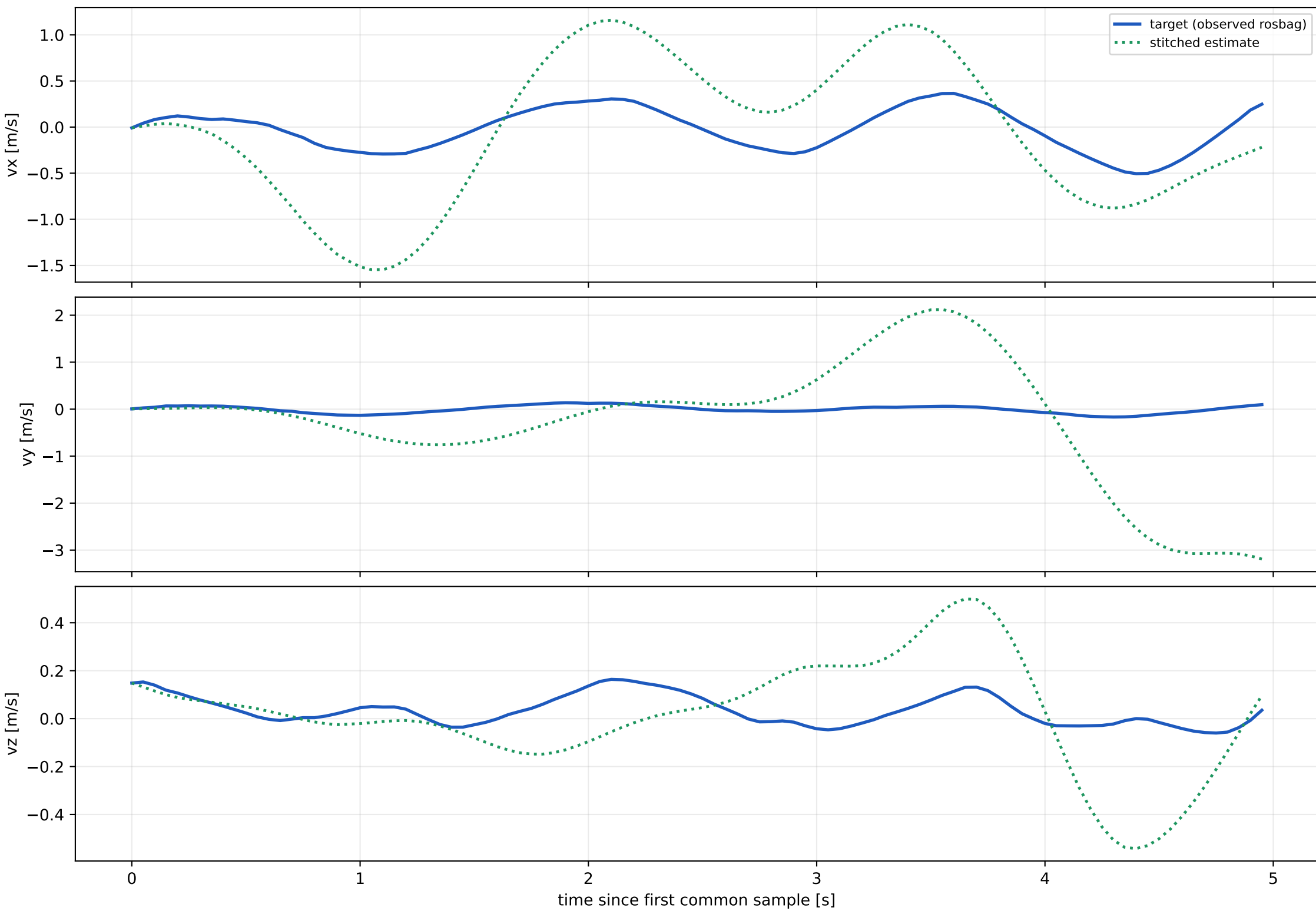
Sensor position



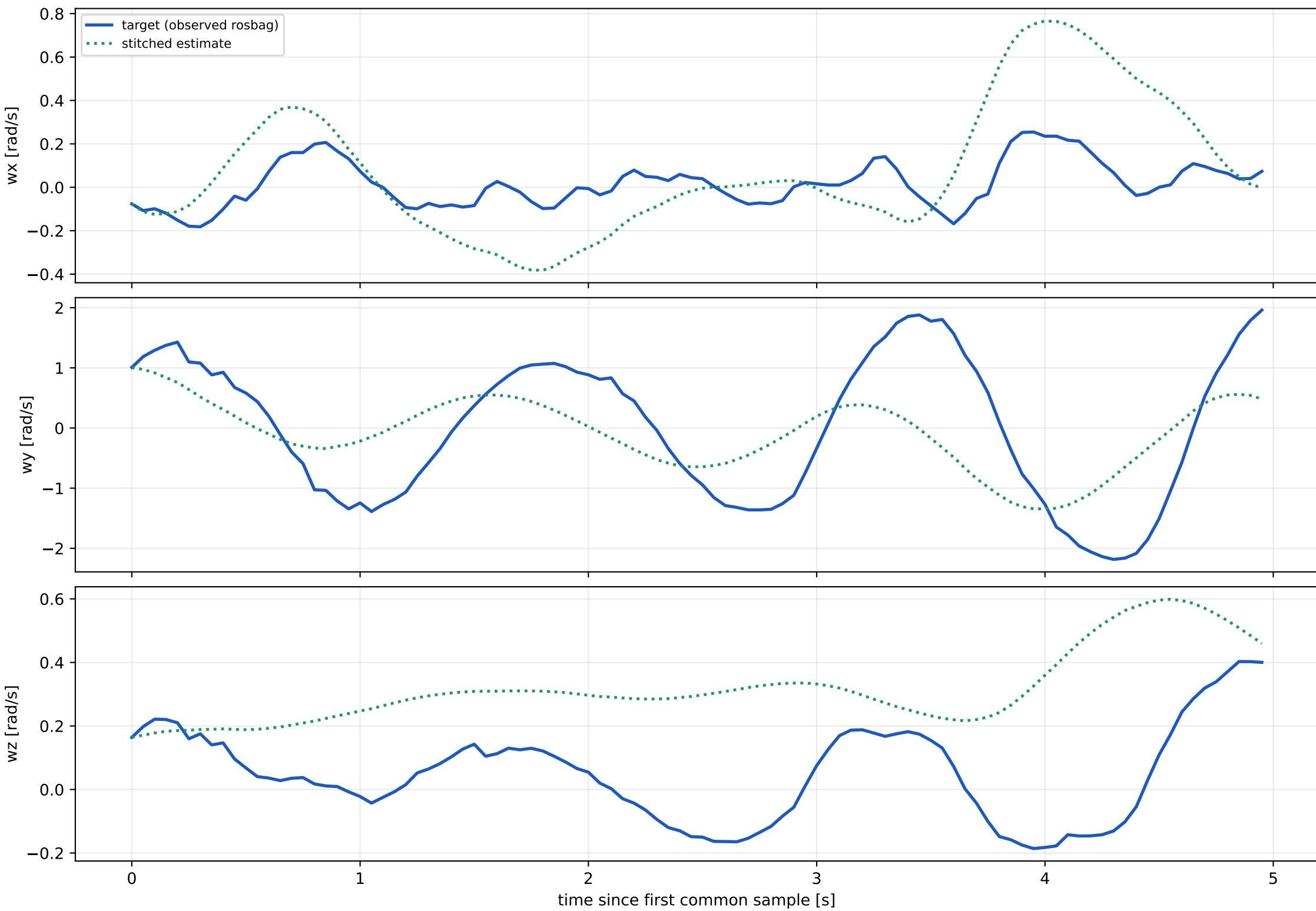
Sensor orientation (display only; fit uses SO(3) log)



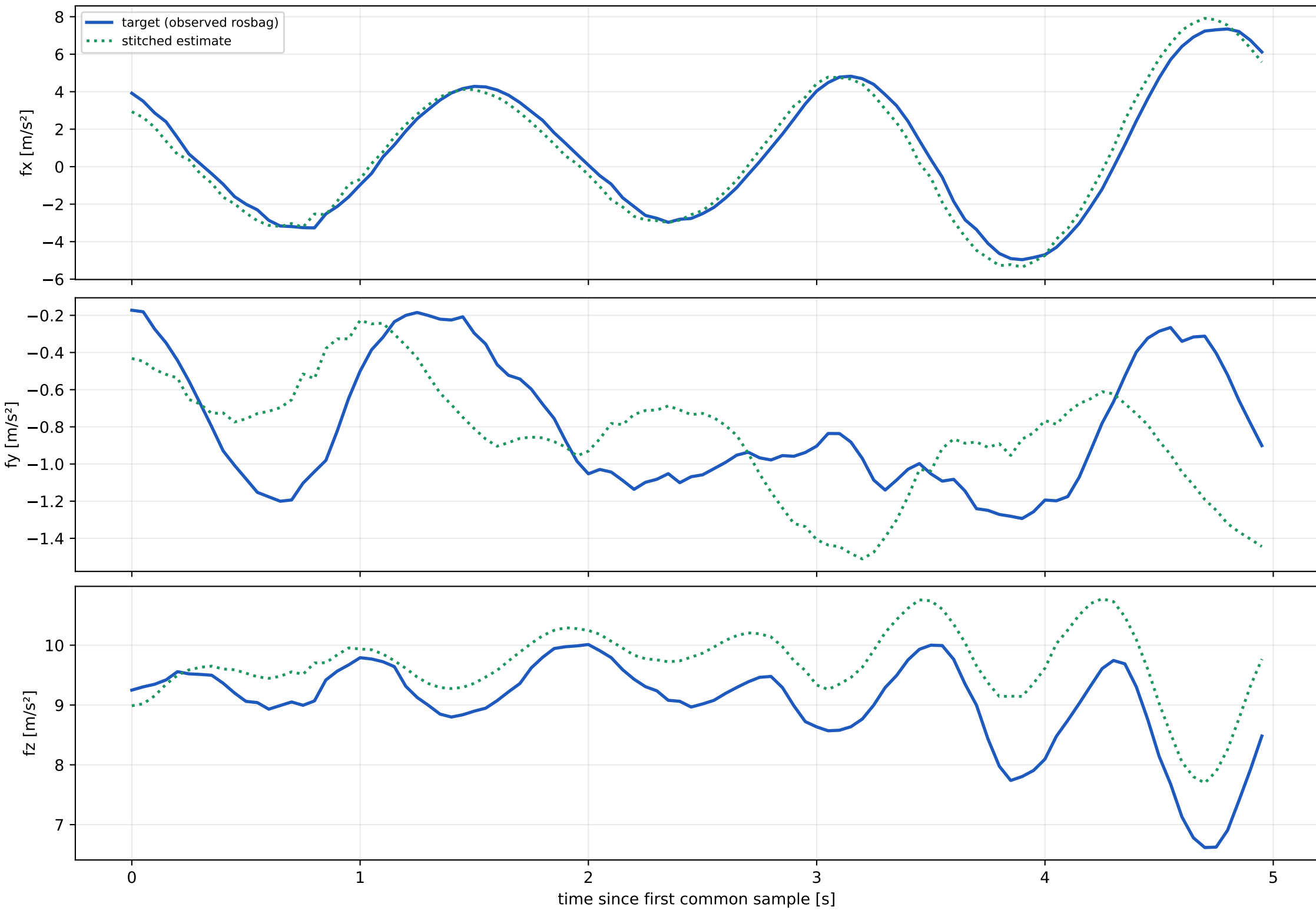
World-frame sensor velocity (diagnostic; excluded from fit)



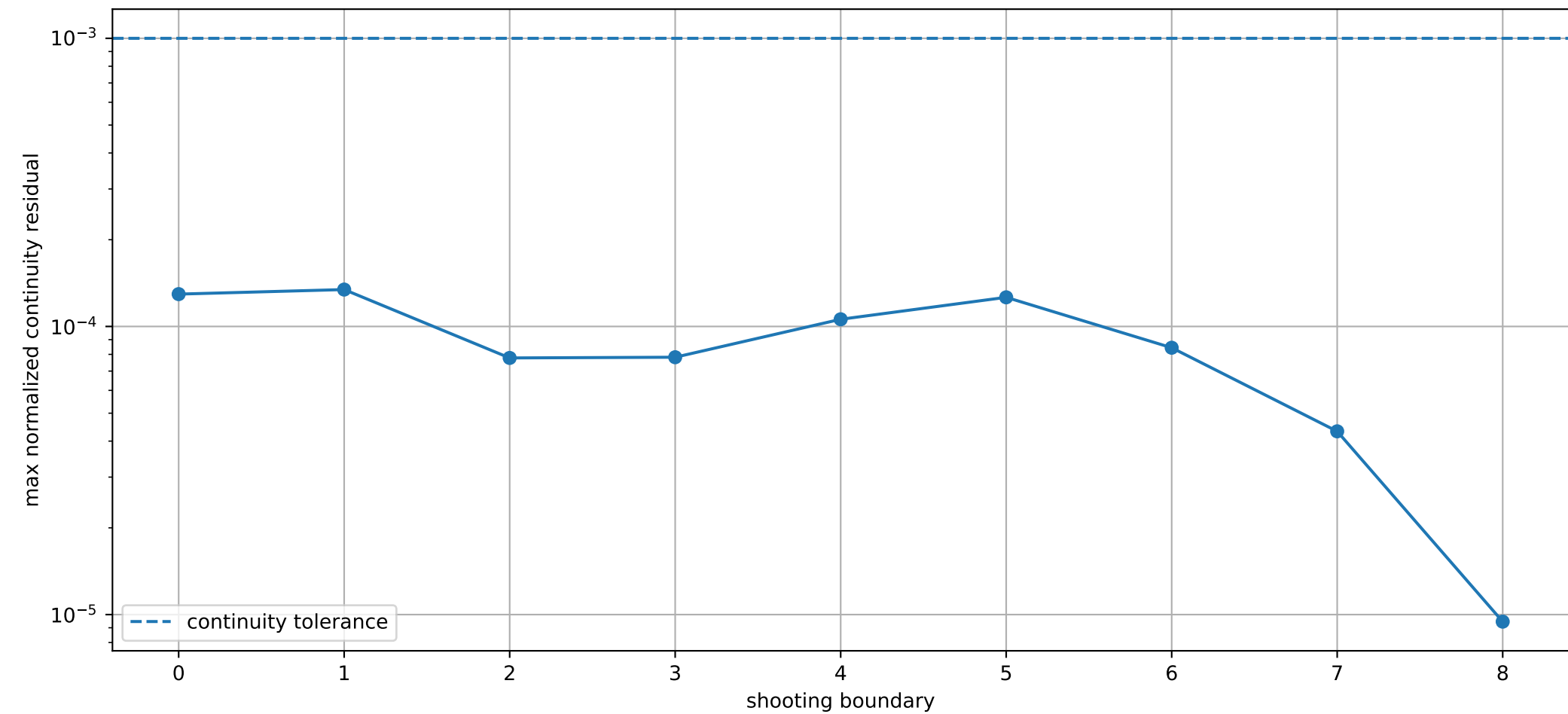
IMU angular velocity (diagnostic; excluded from fit)



IMU specific force (diagnostic; excluded from fit)



Shooting continuity diagnostic



Final estimated parameters

Selected multiple-shooting estimate

Decoded physical parameters

```
mass [kg] 2.59584935708
inertia [kg m^2]
[ 0.0837313036287 -0.00794774236842 -0.00180465388552]
[-0.00794774236842 0.294653865538 -0.000999592122333]
[-0.00180465388552 -0.000999592122333 0.377376117722]
principal inertia [kg m^2] [0.0834207202977, 0.294942406007, 0.377398160584]
CoG offset [m] [0.0225972925888, 0.00917593188377, 0.00945015050417]
rotor force effectiveness [0.943247879376, 1.02428843974, 1.23841182552, 0.835770043874]
command delay [s] 0.05272551227
```

Thirteen optimized smooth coordinates

```
log_mass_scale 0.0988188452428
log_second_moment_cholesky_xx_scale 0.758925552195
log_second_moment_cholesky_yy_scale 0.126007045662
log_second_moment_cholesky_zz_scale 0.00235231810666
normalized_second_moment_cholesky_yx_offset 0.0576910326107
normalized_second_moment_cholesky_zx_offset 0.0456094512929
normalized_second_moment_cholesky_zy_offset 0.0442346579369
cog_offset_x_m 0.0225972925888
cog_offset_y_m 0.00917593188377
cog_offset_z_m 0.00945015050417
force_effectiveness_contrast_1 -0.0582828060265
force_effectiveness_contrast_2 -0.188646451367
force_effectiveness_contrast_3 0.207155321412
```

Fit diagnostics

```
stitched inertia-radius loss [m^2] 0.283404291725
full-rollout inertia-radius loss [m^2] 0.283489813441
broad soft-prior cost 0.158285809668
position RMSE [m] 0.708849948534
orientation RMSE [deg] 48.3413068545
velocity RMSE [m/s] 1.47582526408
angular velocity RMSE 1.02751835581 rad/s
specific force RMSE 1.07472048224 m/s^2
continuity max (normalized) 0.000134283931852
continuity tolerance 0.001
continuity converged True
```